Lab 2: SankofaPOWER – What is GeoAI?

Lab Instructions

In this lab we will build on work from Lab 1 and explore some more computational GIS techniques.

Get More Data

In this section we will explore two ways to acquire geospatial data.

1) Go to https://overpass-turbo.eu/. This is a web-based data filtering tool for OpenStreetMap (OSM) – "a map of the world, created by people like you and free to use under an open license."

a) Below is a query to grab all the information about Oak Park Center, where Project Sweetie Pie does a lot of work in North Minneapolis. Enter this into the console at the top of the screen. Delete everything else!

```
{{geocodeArea:"Minnesota"}}->.State;
way
["name"="Oak Park Center"]
(area.State);
out center;
```

b) Click **Run.** If everything ran correctly, click on the magnifying glass to see the data.

c) We could set up a program to automate this task or grab even more information from OSM, but this will suffice for this lab. Click **Export | GeoJSON | download**

d) **Rename** this file to **OakPark_Center.geojson** and move it to the Data folder from lab 1

2) Next, let's go to https://opendata.minneapolismn.gov/datasets/cityoflakes::pw-street-centerline/about. This is a dataset of all roadways in Minneapolis.

a) Click **Download | GeoJSON | Download**

b) Move the file **PW_Street_Centerline.geojson** to the Data folder from lab 1

Routing in QGIS

In this section, we will utilize the data we have gathered to help answer a hypothetical question, "**How can I get a free dinner tonight?**"

1) Reopen the QGIS project from lab 1 (Sankofa.qgz).

2) Add the new data layers

a) Cick on the **Layer** tab at the top of the screen. Select **Add Layer | Add Vector Layer**

b) Under Source, click on the …, navigate to OakPark_Center.geojson, and select open. Then Click the Add button.

d) Repeat for PW_Street_Centerline.geojson then close the Data Source Manager window.

3) Now let’s do some geoprocessing to find the way to Oak Park!

a) Click on the **Processing** tab at the top of the screen. Select **Toolbox**

b) The **Processing Toolbox** pane should appear on the right. Go to **Network Analysis | Shortest path (point to point)** and double click.

c) This will open up a window for the tool. Here are the parameters you will need to specifically set:

i) **Vector layer representing network:** PW_Street_Centerline

ii) **Path type to calculate:** Fastest

iii) **Start point:** Click on the … and then select anywhere in Minneapolis on the map view

iv) **End point:** Click on the … and then select OakPark_Center on the map view

v) Select **Advanced Parameters**

vi) **Direction Field**: TRAFFIC_DIR

Note: These are attributes for all the roads of Minneapolis.

vii) **Value for <u>both</u> directions**: T

viii) **Default direction:** Forward

ix) **Speed Field:** SPEED_LIM

x) **Default speed:** 25

xi) **Shortest path:** Click on the … and select **Save to File** then navigate to your **Data** folder. Select **GEOJSON files** in the bottom right of the file browser. Lastly, type in **Route_to_OakPark** as the name

d) After double checking all those parameters, click **Run**

e) There’s the fastest route to Oak Park Center (by driving)! You may have to adjust the **symbology** of the new layer to make it more visible.

4) Take a screenshot of the map view.

Save this image to your Google Drive and/or email it to instructors to ensure it doesn’t get lost.

<u>Don’t forget to save your QGIS project!</u>